

Long-Lived Charge Trap Characterization for Rare-Event Searches with Skipper-CCDs

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Skipper Charge-Coupled Devices (Skipper-CCDs) are detectors with sub-eV resolution, allowing them to play a critical role in rare-event searches. A major challenge in experiments using these detectors is understanding and mitigating low-energy backgrounds which impact the single-electron rate (SER). One potential component of the SER could arise from single-electron charge traps caused by lattice defects. In this work we use a new technique for generating uniform charge across the CCD to study single-electron traps in a SENSEI sensor placed underground in the MINOS tunnel at Fermilab. The underground shielding allows us to make use of the pocket-pumping technique to probe traps at high temperatures for long periods of time, enabling sensitivity to traps with emission times $> 10^3$ seconds at normal operating temperatures. This regime was not accessible in previous surface measurements of comparable sensors. We fully characterize a large fraction of the identified traps and use this information to extrapolate their contribution to the single-electron background in SENSEI. Through simulation, we find the characterized population of traps does not significantly impact the single-electron rate. However, upon correcting for measurement inefficiency, we find that a substantial population of long-lived traps may exist. If these unobserved traps exist, they could explain a majority of the observed exposure-dependent single-electron rate in SENSEI.

I. TRAPS AND SINGLE-ELECTRON BACKGROUNDS

Skipper Charge-Coupled-Devices (Skipper-CCDs) are pixelated silicon detectors widely used in rare-event searches [1–4]. They are enhanced versions of CCDs that can perform repeated non-destructive measurements of output charge and consequently achieve sub-electron resolution [5], enabling sensitivity to particles that produce one or a few electrons when they interact with the bulk silicon. This has allowed the SENSEI and DAMIC experiments to achieve world-leading results in the search for light dark matter [6–11]. These experiments are currently limited by low-energy backgrounds, particularly contributions to the single-electron rate (SER). These backgrounds may arise from environmental effects [7, 8, 12–14] or intrinsic processes within the sensor [15, 16], and characterizing them is crucial for maximizing the sensitivity of these and other experiments. One possible source of the SER is charge deferred by traps.

It is well known that defects can occur in the silicon lattice within the buried channel of a CCD from con-

taminants or vacancies that trap and release charge after a characteristic emission time determined by the properties of the defect or contaminant [17–22]. If traps have sufficiently long emission times, they can produce uncorrelated single-electron events, therefore contributing to the SER. Results from the OSCURA collaboration [23, 24] indicated that contaminants in the silicon lattice of Skipper-CCDs could contribute to the measured SER, motivating dedicated studies in SENSEI sensors. More recent measurements performed at LAMBDA Laboratory at the University of Buenos Aires on a CCD fabricated from the same wafer as science-grade SENSEI CCDs [25] suggested that current analysis strategies may already account for charge deferred by these traps. However, because of these measurements were taken at the surface, high rates of environmental and cosmic backgrounds limited the sensitivity to a restricted range of trap lifetimes. In order to probe very long-lived traps (traps that emit electrons several days or even months after the initial trapping) at the usual operating temperature of ~ 135 K, it is necessary to take longer pumping measurements at higher operating temperatures, which

must be done in a shielded underground environment. In this work we describe such a measurement, which we perform with SENSEI CCDs ~ 100 m underground in the MINOS cavern at Fermi National Accelerator Laboratory (FNAL).

The Skipper-CCDs used in this study are made of high-resistivity silicon with an active area of 6144×1024 pixels, each of size $15 \times 15 \mu\text{m}$. Each pixel is covered by three overlapping gates (called “clocks” or “phases”) which are held at different voltages to create a potential well for charge. These voltages are clocked between high and low states in sequence to transfer charge from pixel to pixel. Our CCDs are p-channel devices and therefore charge carriers are holes and not electrons. We refer to these holes as electrons below. During readout charge in pixels of the active region is shifted row-by-row to a serial register and then read out one column at a time in the floating gate amplifier, which performs the Skipper readout step. The CCD has four readout amplifiers that each read out in parallel a “quadrant” of 512 rows by 3072 columns. The CCD was operated in a vacuum chamber equipped with a vacuum pump and a cryocooler, and then read out with a Low-Threshold-Acquisition board (LTA) [26].

In Sec. II we describe the measurement protocol for identifying trap candidates, and in Sec. III we apply selections to remove poor-quality traps and characterize the energy, cross-section, and lifetime of our well-behaved trap population. In Sec. IV we perform a data-driven study to estimate the completeness of the measurement. In Sec. V we detail a full readout simulation using the high-energy event spectrum observed in MINOS to test the impact of the observed trap distribution on the SER. Finally, we conclude in Sec. VI with a summary of the impact of the trap density on single-electron backgrounds in SENSEI.

II. POCKET-PUMPING MEASUREMENTS

In order to search for individual traps in this study we make use of the pocket-pumping technique described in detail in [24, 27]. This method requires an approximately flat field of charge in the active area of the CCD. We do not use an external light source to uniformly illuminate the CCDs used in the study. Instead we make use of a pumping technique described in [28]. In particular we use “pixel-pumping,” where a controlled amount of spurious charge is generated in the active area by clocking the vertical gate voltages to shift charge back and forth from pixel to pixel. In this work we used a voltage swing of 10 V and 100,000 shifts to generate approximately 2000 e^- of charge per pixel, which serves as the flat field of charge for pocket-pumping. After generating this flat field, we perform asymmetric pixel pumps between the three pixel phases with varying phase times. If a trap is present in an edge phase, charge may be captured, released later, and spill into the neighboring pixel;

repeating this cycle for a phase near the trap resonance produces a “dipole” signal relative to the uniform background. We refer to our edge phases as ϕ_1 and ϕ_3 , and ϕ_2 is the phase under which charge is held during an exposure.

The number of charge carriers trapped and released by pocket-pumping manifests as the intensity of the dipole ($I_{t_{ph}}$), which is defined as half the difference in charge between the two pixels. This can be modeled using Shockley-Read-Hall theory [29, 30], where $I_{t_{ph}}$ depends on the probability of capture P_c , the trap “effectiveness” or “depth” D_t , and the emission probability P_e . P_c acts as a scaling factor that varies slightly with temperature, and P_e can be expressed as a function of the emission characteristic time $\tau_e(T)$ [24]. The pumps are asymmetric with a short delay time t_{ph} and a longer delay time $7t_{ph}$ (chosen to maximize the trap intensity and minimize the pumping sequence time; see [24]). This gives:

$$I(t_{ph}) = N_{\text{pumps}} D_t P_c \left(e^{-\frac{t_{ph}}{\tau_e}} - e^{-8\frac{t_{ph}}{\tau_e}} \right) \quad (1)$$

The measurement procedure was as follows: for a given temperature, charge-generating shifts were performed as described above to uniformly illuminate the CCD with charge. Then the pocket-pumping protocol was performed with $N_{\text{pumps}} = 3000$ to obtain well-defined non-saturated dipoles for various t_{ph} . At a given temperature, t_{ph} was varied from approximately 5×10^{-5} s to 1.03 s. Only one skipper sample was taken to reduce readout time. Temperature was varied from [125, 210] K with 18–25 images at each temperature. We specifically took longer t_{ph} measurements at higher temperatures to maximize sensitivity to longer trap emission times. The exact pumping sequence is the same as described in [25].

Dipoles are identified with an algorithm similar to the one described in [25]. For each row of the CCD we subtract the median charge and multiply it with the previous row. If a dipole is present, this computation yields a negative value. We require the magnitude of this intensity to be greater than 3σ , where σ is the noise level of the pixel estimated through median absolute deviation. The median is used here to increase the number of possible candidates; otherwise, outliers could suppress genuine traps.

We then log each such trap candidate for each t_{ph} value for a single temperature and discard traps that only appear in a single image. This then forms a trap candidate catalog containing intensity I per t_{ph} per temperature. Uncertainties on this intensity measurement come from two sources added in quadrature: the first being the Poisson noise of the pixel and the second from a baseline noise estimate. The latter is measured from 4000 randomly chosen trap-free pixel pairs per temperature per quadrant: for each pair, the charge in the same two pixels is subtracted across every image in the t_{ph} scan at that temperature, and σ_{base} is the median across pairs of the resulting per-pair standard deviation, divided by $\sqrt{2}$ to

recover the single-read noise floor (varies from 30–40 e^- at colder temperatures to $\sim 80 e^-$ at higher temperatures). We find 9333 trap candidates.

III. TRAP CHARACTERIZATION

The trap candidates for a given temperature are then fit with Eq. (1) plus a pedestal term I_0 . This accounts for a t_{ph} -independent dipole component that we observe at the trap coordinates in data, which grows from a negligible level at 150 K to $\mathcal{O}(700) e^-$ above 175 K. Inspection of the raw images shows that it corresponds to a charge deficit in the trap pixel with the compensating charge deferred along the readout direction, while the summed charge of the pixel pair remains at the background level; it is therefore not a hot (generating) pixel but deferred charge produced by the trap itself acting on the dark-current background as charge packets are clocked through its physical location during readout. Because the readout sequence is identical in every image of a t_{ph} scan, this component is constant across the scan and is cleanly absorbed by I_0 , leaving the pumped term unbiased. Omitting I_0 causes the χ^2 goodness-of-fit test to reject bright trap curves above ~ 165 K, where the pedestal exceeds the per-point noise.

We then select trap candidates with a fitted pump strength ≥ 3 times the fit uncertainty. We evaluate the goodness of fit using a χ^2 test statistic. To select a good-quality trap we require $\Delta\chi^2$, defined as $\chi^2_{\text{flat fit}} - \chi^2_{\text{intensity fit}}$ to exceed a temperature-dependent threshold chosen such that 0.1% of pure noise locations would pass this selection. This threshold is determined with a calibrated detection table built by sampling thousands of trap-free pixel pairs (~ 2000 per quadrant), running the identical trap intensity fits and then computing $\Delta\chi^2$ for every temperature.

This fit provides $\tau_e(T)$ for every trap at a given temperature. To identify traps that follow our model, we perform these fits for a trap candidate at multiple temperatures. If a trap does not have good intensity fits for four different temperatures, it is discarded, resulting in a reduction of trap candidates to 4711. Traps that are lost in the intensity fit process are rejected mostly due to not following the model (flat at all t_{ph}) or having a maximum intensity out of range of the t_{ph} measurement range. We also notice traps (~ 400) that switch orientation and have opposite sign intensity fits for some but not all temperatures, which are indicative of dual-response traps. We discard these traps, and this reduces the candidate count to 4304 traps.

For a given charge trap we model the hole emission time τ_e as a function of lattice temperature T , the thermal velocity of the charge carriers v_{th} , the effective density of states in the valence band N_v , the trap depth above the valence-band edge $E_t - E_v$, and the trap cross-section σ [24, 25, 29, 31]. Defining $E_{\text{trap}} \equiv E_t - E_v$, this gives:

$$\tau_e = \frac{1}{\sigma v_{th} N_v} e^{\frac{E_{\text{trap}}}{k_B T}}, \quad (2)$$

where the temperature-dependent v_{th} and N_v are defined as

$$v_{th} = \sqrt{\frac{3k_B T}{m_{\text{cond}}}} \quad \text{and} \quad N_v = 2 \left[2\pi m_{\text{dens}} \frac{k_B T}{h^2} \right]^{3/2}. \quad (3)$$

Here, h is Planck's constant, $m_{\text{cond}} \simeq 0.41 m_e$ and $m_{\text{dens}} \simeq 0.94 m_e$ are the hole effective masses for conductivity and density of states, respectively, between 100 K and 200 K [32], and m_e is the electron rest mass.

For each trap candidate, we track the intensity fits across all temperatures. We then fit these measurements with Eq. (2), and reject traps that have a reduced $\chi^2 > 10$. This results in a final characterized trap count of 3798 traps. An example is shown in Fig. 1.

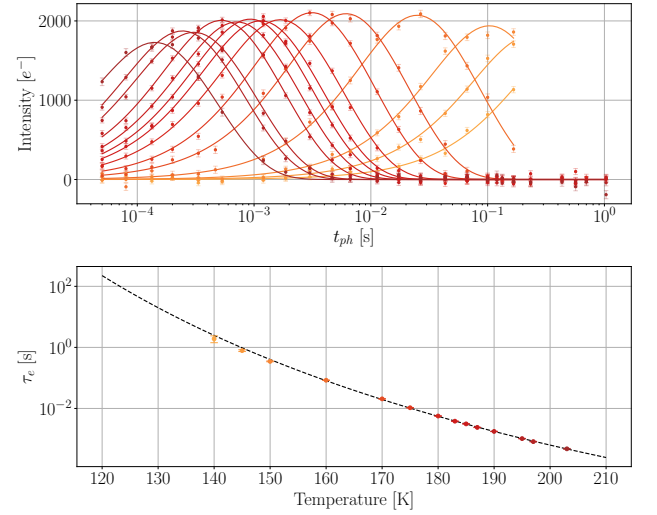


Figure 1. An example of a characterized charge trap in quadrant 1 of our CCD is shown. The **top** panel shows the intensity (in units of electron charge) as a function of t_{ph} fitted according to Eq. (1). The **bottom** panel shows τ_e as a function of temperature for the same trap with the fit line from Eq. (2). The points are colored by temperature. Here the pedestal has been removed and the magnitude of the intensity is plotted.

We analyze the spatial distribution of these identified traps using toy Monte-Carlo simulations. 10^4 simulations were performed under the assumption of spatial uniformity. Fig. 2 shows the histogram of the distances between all identified traps per quadrant, along with the simulation result and its corresponding 90% confidence band assuming the traps are distributed uniformly. Some deviation from uniformity is expected due to the spatially localized nature of the source of traps (defects in the CCD, high-energy events, etc.), but in general the characterized trap population is uniformly distributed.

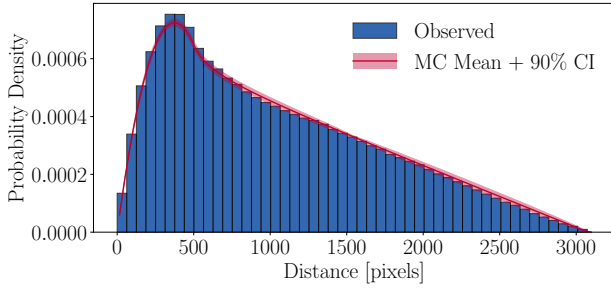


Figure 2. Probability density of the distance between all characterized traps, combining all four quadrants. The red curve and shaded band show the mean and 90% confidence interval from 10^4 toy Monte-Carlo simulations assuming a spatially uniform trap distribution. We see good agreement between the simulation and the data.

We then calculate the energy and cross-section of the characterized trap population, which is presented in Fig. 3. The shaded gray region begins where we no longer have direct measurements of τ_e for any trap; all lines in this region are extrapolated from fitting with Eq. (2). The white region denotes the measured trap emission times. The longer delay times and high-temperature measurements done underground at MINOS allow us to probe significantly higher energies and cross-sections than were accessible previously [25]. We observe a large population centered around 0.2 eV and a smaller secondary population centered around 0.4–0.5 eV, which we dub the “high-energy” population. The latter population was not previously accessible with the pocket-pumping technique.

We then use Eq. (2) to extrapolate the characteristic emission times for all characterized traps at 135 K, the typical operating temperature of SENSEI@SNOLAB [8]. Not all traps have direct τ_e measurements at every temperature.

IV. MEASUREMENT COMPLETENESS

At temperature T , a trap’s emission time is classified as *measured* if the intensity was successfully fit at that temperature and as *extrapolated* if the emission time is obtained solely from the fit to Eq. (2). To account for possible inefficiencies in measuring a given trap and to account for populations of traps that were not characterized at any temperature, we perform a detailed data-driven toy Monte-Carlo study.

First, noise samples were built using the trap-free portions of the images in the data set for each temperature and clock time t_{ph} . The pedestal term I_0 was also included by drawing from the measured per temperature distribution. Next, fake intensity curves following Eq. (1) were built using a grid of amplitudes, where the amplitude is defined as $A = N_{\text{pumps}} D_t P_c$. The grid purposefully spans over a range far beyond the range ob-

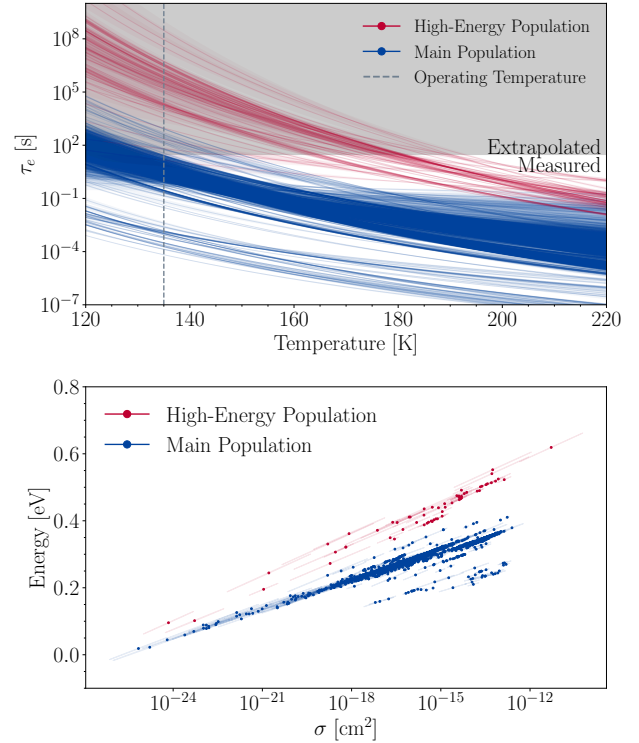


Figure 3. All 3798 characterized traps, colored by observational group: High Energy (red; measured above 170 K) and Main Population (blue; measured between 120–170 K). **Top:** Emission time τ_e as a function of temperature. Solid portions of each line indicate temperatures at which the trap was directly measured; the shaded gray region marks the extrapolated regime where no trap has a direct measurement at any temperature. The dashed vertical line marks the SENSEI@SNOLAB operating temperature of 135 K. **Bottom:** Fitted trap energy E_t versus capture cross-section σ for the same traps.

served in data (200–15,000 e^-). These intensity curves were then injected into the noise samples with the corresponding uncertainties determined by the same procedure described in Sec. II. This allowed us to reproduce the signal structure and fluctuations of real intensity curves for a synthetic set. We then combine the τ_e grid with an energy grid (0.04–0.7 eV) to form a 2D set. For a given set point we can use Eq. (2) to obtain τ_e at all temperatures. We then interpolate the injection-recovery grid to determine $p_{\text{det}}(T, \tau_e(T), A)$. For a fixed amplitude we combine those 23 unequal probabilities with a Poisson-binomial tail. $P(\text{at least 4 good fits}) = P(\sum_{i=1}^{23} \text{Bernoulli}(p_i) \geq 4)$. We then marginalize this over the empirical trap amplitude. This is done by factoring out the N_{pumps} term and then characterizing the trap depth D_t and P_c for all temperatures. D_t ranges from 0.2 to 1.2 and is independent of temperature. P_c is temperature-dependent; it is close to ~ 1 for $T < 180$ K but drops logarithmically to 0.78 at 210 K.

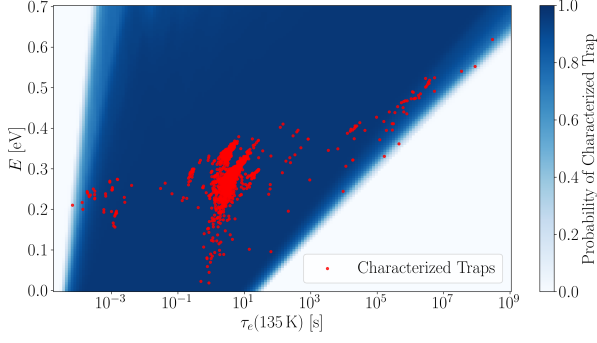


Figure 4. Results of the data-driven sensitivity study are plotted alongside the actual energies and emission time of the characterized traps. The darker colors correspond to high detection probability, and the lighter colors show where this measurement is not sensitive. As expected, most of the characterized traps fall in the high probability region.

This amplitude is averaged over D_t and is scaled by the temperature-dependent P_c . This results in a 2-D probability distribution $P(\text{characterized} | \tau_{135}, E)$, shown in Fig. 4. We validated this grid against our characterized population and found that 99.79% of characterized traps lie in regions with $P_{\text{characterized}} > 0.8$.

Interestingly, the presence of traps near the low-probability border implies a hidden population of long-lived traps much larger than the observed set. We average this probability distribution over the energy distribution of observed traps under the assumption that the hidden trap population has an energy distribution similar to the observed population. This collapses the measurement probability into an effective efficiency of measurement as a function of trap emission time. This is shown in Fig. 5 along with the observed distribution of τ_e at the SENSEI operating temperature of 135 K. This demonstrates that traps are only reliably characterized within $10^{-3} \text{ s} \lesssim \tau_e \lesssim 10^4 \text{ s}$. It is worth emphasizing that this efficiency assumes that hidden traps have energy and amplitude distributions similar to those actually characterized, and makes no statement about traps that do not follow our model. If one were to relax this assumption, for example, by positing hidden traps are twice as faint, then the missed trap population would be inflated by a factor ~ 2 .

V. SIMULATING TRAP EMISSION

To investigate the effect of these traps on our single-electron rate, an end-to-end simulation architecture capable of simulating SENSEI conditions was developed. In the simulation, a “CCD” is initialized with a given charge trap density chosen to be the same as the characterized trap population (3798 traps / 6144×1024 pixels) and an exposure-dependent and exposure-independent single-electron rate taken from the MINOS UR quadrant in pre-

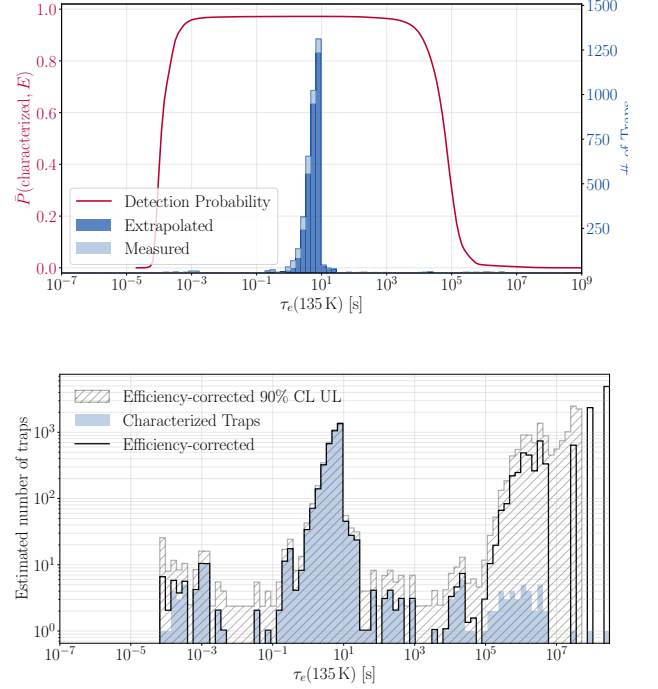


Figure 5. **Top:** Histogram of the predicted emission time τ_e at 135 K for all 3798 well-characterized traps. Green bars indicate traps for which τ_e was directly measured; blue bars indicate traps extrapolated using Eq. (2). The probability of characterizing a trap at four temperatures for a given τ_e averaged over the observed energy distribution of characterized traps is shown in red. This defines an approximate measurement efficiency under the energy and amplitude assumptions discussed in the text. **Bottom:** The same trap distribution overlaid with the efficiency-corrected histogram in black and the efficiency-corrected Poisson 90% upper-limit histogram in hatched lines. This predicts a substantial population of long-lived traps.

vious SENSEI results [14]. The exposure-dependent rate is taken as $4.36 \times 10^{-5} e^-/\text{pix}/\text{day}$ and the exposure-independent rate is taken as $9.94 \times 10^{-5} e^-/\text{pix}/\text{image}$. For simplicity, only one quadrant is simulated (3072 columns $\times$ 512 rows). The traps are then randomly placed throughout the quadrant with emission times drawn from the distribution of the characterized traps shown in Fig. 5. We assume an equal distribution of traps under ϕ_1 and ϕ_3 , all initialized as empty. Traps under the ϕ_2 phase were not measured or simulated.

The simulation takes “images” by specifying an exposure time and injecting a number of single-electron counts randomly throughout the image based on Poisson fluctuations around the exposure-dependent and exposure-independent single-electron rates. High-energy events must also be simulated in order to match science conditions and provide the mechanism for filling traps. To simulate data-taking conditions, we sample a fraction of high-energy events (defined by choosing contiguous regions of positive charge with total charge $> 100e^-$) from

a 20 h SENSEI image taken at MINOS. We ignore long horizontal streaks as these correspond to high-energy events that strike the serial register during readout. For a chosen cluster we transplant all charge within a 60-pixel radius of pixels with charge $> 100e^-$ into random non-overlapping positions in the image. The number of high-energy events inserted in the image scales linearly with exposure; e.g. for a 4 h image, a random fifth of the total high-energy events are chosen. To approximate SNOLAB conditions we scale down the number of high-energy events by a factor of 10 based on the high-energy background rate measured in [14].

We then simulate readout row-by-row with delay times corresponding to two different data-taking procedures. The first simulates 32×1 binning as in SNOLAB Run 2 [14] (~ 13 min readout), and the second assumes no binning as in SNOLAB Run 1 [8] (~ 7 h readout). Since we only model traps under the edge phase, traps can only capture charge during the phase overlap $dt \sim 20 \mu\text{s}$ of a row shift with probability $P_c = 1 - e^{-dtq\sigma v_{th}/V_p}$ where q is the packet occupancy, σ the trap cross-section (sampled jointly with τ_e from the measured population), and V_{p432} the effective volume explored by a single carrier confined in the pixel well. Filled traps then emit with probability $P_e = 1 - e^{-\Delta t/\tau_e}$. After an image is read out, we simulate the CCD clear used in operating conditions. We evaluate the emission probability after each row transfer dwell during readout/clearing and after every exposure. If emission occurs during vertical transfer we allow traps under ϕ_3 only to recapture the emitted electron with the same probability described earlier.

The effective packet volume V_p is estimated using the product of a single carrier's lateral and vertical extent inside the potential well of the pixel. The lateral active-channel width between parallel channel stops is given by multiplying the channel-stop width $12 \mu\text{m}$ [28] by one-third of the pixel pitch ($5 \mu\text{m}$) as the three phases are equal width [2]. The vertical extent is estimated by the thermal (Boltzmann) spread of a carrier in the buried-channel potential well $\sigma_z = \sqrt{\epsilon_S k_B T / (q^2 N_A)} \approx 24 \text{ nm}$ using a channel doping $N_A \approx 1.3 \times 10^{16} \text{ cm}^{-3}$ obtained from the implant dose [2] and buried-channel depth [28]. This gives $V_p \approx 1 \mu\text{m}^3$, which we use as the representative value for our simulation.

A single trial refers to a single initialized CCD state in which 500 paired images (with and without traps, taken simultaneously) are taken and read out with shuffled exposure times of 0 h, 4 h, 6 h, 10 h, and 20 h. After every image is read out, we simulate a 3 h clear as was used in prior science runs [8, 14]. After the images are taken we apply a set of masks developed in previous analyses [7, 8, 14]. We specifically apply the halo mask (60-pixel radius around a pixel with $> 100e^-$), vertical bleed mask (all pixels above a pixel with $> 100e^-$ are masked), and the hot column and hot pixel masks. For the hot column mask, which is statistical, an A/B strategy is implemented, where half the simulation set is designated as set A, and the other half as set B. The set B statistics

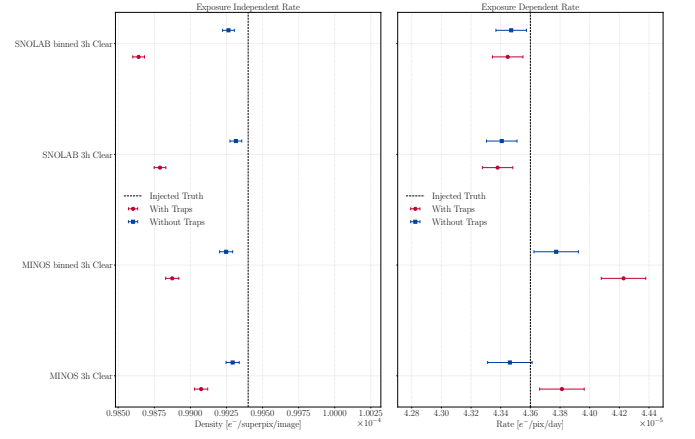


Figure 6. Exposure-independent (left) and exposure-dependent (right) rates for no-trap (blue) and trap (red) images for MINOS and SNOLAB science data-taking conditions. For the characterized population, differences are of order $\sim 10^{-7}$, and therefore do not impact the SER.

are used to decide the hot columns and pixels in A, and vice versa. This avoids truncation bias. We then perform 200 such trials for binned and unbinned readout for both MINOS and SNOLAB conditions (800 total trials).

After masking, the single-electron densities are calculated for all exposures. They are then fit to obtain exposure-independent and exposure-dependent rates for images taken with and without traps. The results are shown in Fig. 6. We see the no-trap branch recovers injected truth of the exposure-independent charge $< 0.3\%$ and the exposure-dependent charge to within $< 1.3\%$, providing a reasonable validation of the framework. We see that if traps match the characterized population then the effect on the SER is of order 10^{-7} and therefore is negligible compared to the single-electron rate. For these images we see that the exposure-independent charge with traps is lower than the injected truth; this is because single-electron events rarely get trapped and are lost when they emit during the clear. These results demonstrate that masking is effective for traps with relatively short emission times (< 100 s), which constitute a majority of the characterized traps.

However, the results from the efficiency study imply a significant population of unmeasured traps with very long emission times. To test the effect on the SER we ran the same simulation set on two populations. The first was the efficiency-corrected trap population, calculated by dividing the trap count by the measurement efficiency plotted in Fig. 5. The second was obtained by taking a Poisson 90% confidence level upper limit on the trap counts, and then applying the same efficiency correction. To prevent small weights from spuriously increasing the counts of the upper-limit-corrected population we restricted the bounds of the efficiency correction from $\tau_e \in [6 \times 10^{-5}, 5 \times 10^7]$. The lower edge was defined by the extent of the completeness band and the upper edge was defined by requiring efficiency $\epsilon > 0.001$. We

note that traps with emission times faster than the lower bound are irrelevant for masking as they emit nearly instantaneously. For the upper bound we tested increasing the upper-limit correction to 10^8 s and found the SER does not shift significantly; this is because these traps are too long-lived to contribute and largely act as sinks. When simulating the efficiency-corrected and upper-limit populations we increased the trap densities from the characterized trap density of 6.04×10^{-4} to 2.34×10^{-3} and 2.86×10^{-3} respectively to account for the increase in counts.

To test the effect of these traps on the exposure-dependent rate, we ran the same simulation pipeline without injecting exposure-dependent charge for both conditions. We draw a band between the efficiency-corrected population and the upper-limit population to define a plausible range for the exposure-dependent trap-driven charge excess. As seen in Fig. 7, a hidden trap population could explain $\sim 25\%$ – 55% of the exposure-dependent SER in MINOS. At SNOLAB we see that the lower boundary is close to 40% and the upper boundary exceeds that of the measured exposure-dependent SER at SNOLAB. MINOS is closer to the surface and has a higher background rate, so it is plausible the measured SER is a composite of charge traps and environmental backgrounds. SNOLAB has a significantly lower rate of environmental backgrounds, and so traps could make up a larger fraction. We note that current masking procedures seem to be ineffective for these long-lived trap populations.

There are several possible sources of uncertainty on these results. The trap-driven excess arises from a relatively small number of traps, introducing a statistical uncertainty of 22%. Overall systematic uncertainties include the choice of V_p , the two assumptions that hidden traps would have similar amplitude and energy distributions to the characterized population, and the choice to sample high-energy events from a single MINOS image. The first was checked; varying V_p by an order of magnitude in either direction results in shifts of order $\sim 10^{-7}$ in this SER, and so the systematic uncertainty is relatively small. For the second, as mentioned earlier, if hidden traps had fainter amplitudes then the hidden population would be biased larger; however, this would not hold for brighter amplitudes as these would be easier to detect. We note the energy range assumption is supported by the traps characterized in the literature [18, 24, 33, 34].

VI. SUMMARY AND OUTLOOK

In this work we characterized charge traps in a science-grade SENSEI Skipper-CCD operated underground in the MINOS cavern at Fermilab. Using the pocket-pumping measurement technique performed at 23 temperatures ranging from 125–210 K, we identified 3798 traps in the sensor. These measurements allowed us to probe trap emission times of hundreds of days

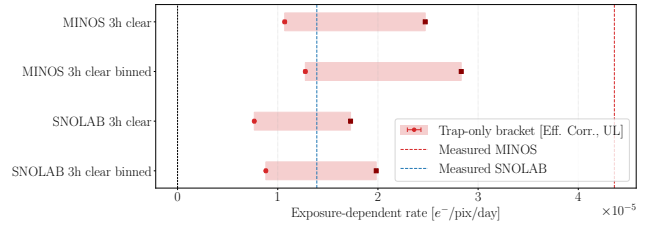


Figure 7. The band between the efficiency-corrected population (Eff. Corr) and the efficiency-corrected 90% CL upper limit histogram is shown for the same data-taking conditions as SENSEI@SNOLAB and SENSEI@MINOS. We show the two populations to highlight the possible excess in the exposure-driven SER attributable to charge traps. The best-measured rate at MINOS (injected in the original simulation) is shown as a vertical red dashed line, and the golden quadrant result from [14] is shown as a blue vertical dashed line. This implies the effect of the traps on the SER could be significant, potentially explaining all of the exposure-dependent SER for SENSEI@SNOLAB.

the SENSEI@SNOLAB operating temperature of 135 K, a regime previously inaccessible.

The characterized traps cluster predominantly at $\tau_e \sim 10^{-1}$ – 10^2 s at 135 K, with a tail extending to multi-day lifetimes. To bound the contribution of traps that fall outside our measurement window at all temperatures, we introduced a measurement efficiency $\varepsilon(\tau)$ through a data-driven study of measurement completeness. We obtained two efficiency-corrected trap populations, one by reweighting the characterized trap distribution and the other by applying the same procedure after taking the 90% CL upper limit. This provided a quantitative bound on the contribution of the uncharacterized population that complements the direct characterization.

We tested the effects of these populations on the SER with a dedicated CCD simulation by injecting exposure-dependent and exposure-independent rates from the best MINOS quadrant in [14]. The directly characterized population has a very small effect on the SER, producing excesses of order 10^{-7} . We also tested the efficiency-corrected and upper-limit populations in simulation without injecting any exposure-dependent rate. This hidden population significantly increases the SER and can explain a large fraction of the observed SER in MINOS and SNOLAB.

These results demonstrate the value of characterizing charge traps for single-electron-sensitive experiments, and will be especially relevant for upcoming large-scale efforts such as DAMIC-M and Oscura. The spurious-charge illumination technique demonstrated here is well-suited to characterizing traps in all sensors of such experiments simultaneously, without dedicated test stands. Since the trap population is expected to depend on fabrication, repeating these measurements on each new sensor batch will be important. These results suggest, but do not prove, that a large fraction of the exposure-dependent single-electron rate could arise from a popu-

lation of charge traps with very long emission times. To test this conclusively, new techniques for probing these very long-lived traps must be explored.

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